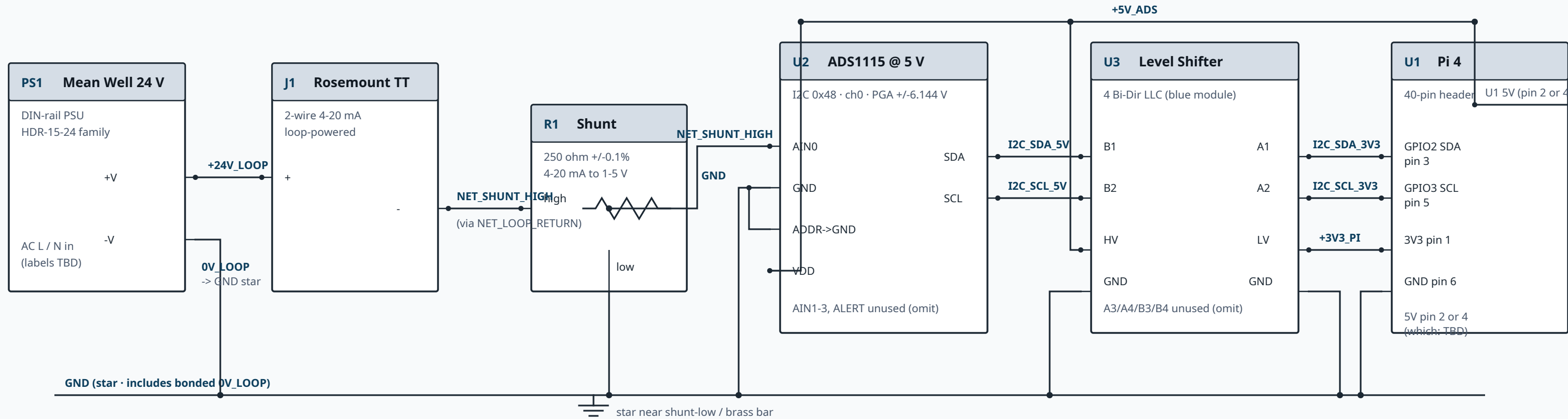


DCS - Process Loop & ADC Interface

Block schematic · electrical only · source: docs/hardware/CONNECTIONS.md (rev D)

Flow left to right: PS1 24 V □ Rosemount □ R1 250 Ω □ U2 ADS1115 □ U3 LLC □ U1 Pi 4 · 0V_LOOP bonded to GND at star · unused pins omitted (not invented)



Designator map (CONNECTIONS.md names)
PS1 <- PSU_MW Mean Well DIN 24 V
J1 <- TT_RM Rosemount 2-wire TT terminals
R1 <- RS250 250 ohm shunt (+/-0.1%)
U2 <- U_ADS ADS1115 breakout @ 5 V, ADDR->GND = 0x48
U3 <- U_LLC 4 Bi-Directional Level Shifters
U1 <- U_PI Raspberry Pi 4 Model B
Omitted (not permanent / unverified): Fluke 789, pump relay GPIO17, WebSocket / UI software. Unused pins omitted (AIN1-3, LLC A3/A4/B3/B4, ALERT) - no invented GND ties.

Confirmed loop math
$V = I \times 250 \text{ ohm} \cdot 4 \text{ mA} \rightarrow 1.000 \text{ V} \cdot 20 \text{ mA} \rightarrow 5.000 \text{ V}$ 8.00 mA -> 2.000 V -> 25% span (bench / Fluke checkout)
Series loop: PS1 +V -> J1+ -> J1- -> R1 -> GND / PS1 -V
Sense: NET_SHUNT_HIGH -> U2 AIN0 (vs GND)
I2C: U1 GPIO2/3 <-> U3 A1/A2 <-> U3 B1/B2 <-> U2 SDA/SCL
PE / brass ground bar lands on same GND star (CONNECTIONS §1). No optional AIN0 clamp shown - not in CONNECTIONS permanent BOM.

PROJECT		DCS - Distributed Control System (bench)	
TITLE		Block schematic - loop, shunt, ADS1115, LLC, Pi	
DWG SCH-DCS-003		REV E	SHEET 1 / 1
DATE 2026-09-28		SCALE NTS	SIZE A3L
Truth: CONNECTIONS.md rev D · No company branding Human-readable drawing - not the SKIDL auto-layout .kicad_sch			